Statistical Thinking and Describing Data

APHS 20203

How Much Are Students Sleeping?

12 of the 150 responses from a fictional TCU survey. Each row represents one student.

Respondent	Sleep hours per night	Respondent	Sleep hours per night
001	5.0	007	4.5
002	7.0	008	7.5
003	6.5	009	6.0
004	5.5	010	5.0
005	8.0	011	6.5
006	6.0	012	7.0

Average Sleep Duration

Across **all 150 respondents** in the fictional TCU survey, the average is **6.2 hours per night**.

What conclusions can the group draw from these data?

What Statistics Adds

Question	What we need
What did we observe?	Counts, percentages, summaries, and graphs
How much do observations differ?	Measures of spread
What might this tell us about a population?	Sampling information and statistical inference

Target Population, Source Population, and Sample

Suppose the survey recruits students leaving the library one evening.

Group	Who belongs to it?	In our example
Target population	People we want findings to apply to	All students enrolled at TCU in Fall 2026
Source population	People from whom we select participants	TCU students leaving that library that evening
Study sample	People whose data we analyze	The 150 students who respond to the survey

How well might these respondents represent all TCU students?

Descriptive and Inferential Statistics

Descriptive Statistics

Describe the **sample we observed**.

Inferential Statistics

Use sample data to make inferences about the **target population**.

Question: Describing or Inferring?

Statement A

“The 150 respondents reported an average of 6.2 hours of sleep.”

Statement B

“We estimate the average sleep duration of all TCU students from the sample.”

1. Which statement describes observed data? Which makes an inference?
2. What would you need to know before trusting Statement B?

Population and Sample Summaries

We use the mean sleep duration in our **sample** to estimate the mean in our **target population**.

What do we call a numerical summary of a population? Of a sample?

	Target population	**Sample**
Numerical summary	**Parameter**	**Statistic**
Example	Mean sleep for all TCU students	Mean sleep for 150 respondents
Mean symbol	μ	$\overline{x}$

A sample statistic can estimate a population parameter.

Question: Statistic or Parameter?

The mean sleep duration among 150 surveyed TCU students is 6.2 hours. What is 6.2 hours?

A. A population parameter

B. A sample statistic

C. Both

D. Neither

What Is a Variable?

How would you define a variable? Give one example from a student health survey.

A **variable** is a characteristic that can take different values across observations.

One student	Values recorded
Sleep duration	6.5 hours per night
High stress reported	Yes
Stress rating	High
Number of caffeinated drinks yesterday	2

Review: Categorical Variables

What are the two types of categorical variables? How do they differ?

Kind	Distinction	Example
Nominal	Categories without a natural order	Major; yes/no response
Ordinal	Categories with an order	Low/moderate/high stress

A numerical code does not automatically make a category quantitative.

Review: Quantitative Variables

What are the two types of quantitative variables? How do they differ?

Kind	Distinction	Example
Discrete quantitative	Countable numerical values	Number of drinks
Continuous quantitative	Measurements along a continuum	Sleep duration

Classify the Recorded Values

Classify each variable.

1. Caffeine use: **yes / no**
2. Sleep duration: **5.5 / 6.0 / 7.5 hours**
3. Stress rating: **low / moderate / high**
4. Number of nights studied past midnight: **0 / 1 / 2 / … / 7**

Choose a Summary

Choose the summary from the variable and the question.

Variable	Start with
High stress: yes/no	Counts, percentages, bar graph
Low/moderate/high stress	Ordered counts and percentages
Sleep hours	Distribution, center, and spread

A Small Practice Dataset

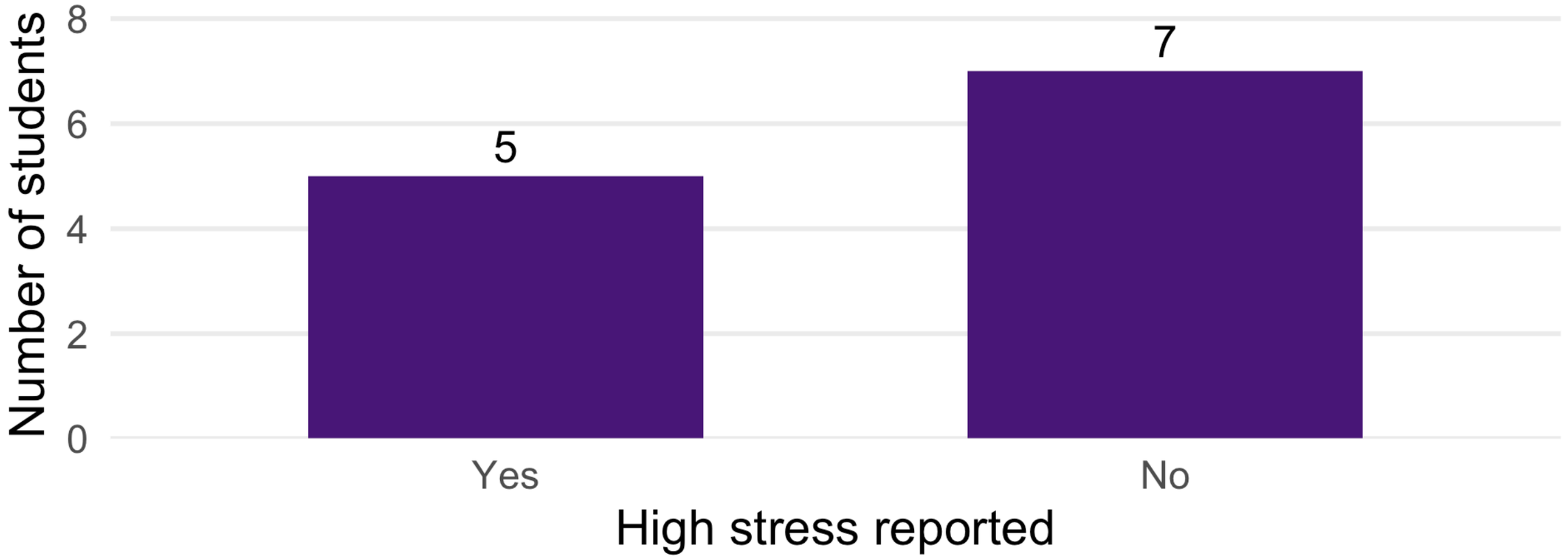
12 fictional students. Each row represents one student.

Student	Sleep hours	High stress?	Student	Sleep hours	High stress?
S01	5.0	Yes	S07	7.0	Yes
S02	5.5	Yes	S08	7.0	No
S03	6.0	Yes	S09	7.5	No
S04	6.0	No	S10	8.0	No
S05	6.5	Yes	S11	8.0	No
S06	6.5	No	S12	9.0	No

Counts and Percentages

High stress reported	Count	Percentage
Yes	5	41.7%
No	7	58.3%
Total	12	100%

A Bar Graph of the Same Responses



Grouped Frequency Tables

Sleep duration	Count
5 to less than 6 hours	2
6 to less than 7 hours	4
7 to less than 8 hours	3
8 to less than 9 hours	2
9 to less than 10 hours	1
Total	12

Where does someone reporting exactly 7 hours belong?

What Grouping Hides

The **7 to less than 8 hours** interval contains **3 students**.

From that grouped table alone:

- We know there are three observations in the interval.
- We do **not** know how many are exactly 7 hours.
- We cannot recover the exact mean sleep duration.

A Familiar 2×2 Table

Sleep duration	High stress: yes	High stress: no	Total
Less than 7 hours	4	2	6
7 hours or more	1	5	6
Total	5	7	12

Which group has the larger percentage reporting high stress?

Question: Which Denominator?

Of 12 fictional students, 6 report less than 7 hours of sleep. Of those 6, 4 report high stress. What percentage of the shorter-sleep group reports high stress?

A. 4 / 12 = 33.3%

B. 6 / 12 = 50.0%

C. 4 / 6 = 66.7%

D. 6 / 4 = 150.0%

Group Exercise: Describe This Sample

In groups of 3–4, use the data on the next slide.

1. Make a two-row frequency table for **high stress: yes/no**. Include counts and percentages.
2. Calculate the percentage sleeping **less than 7 hours within each stress group**: high stress yes; high stress no.
3. Write one sentence describing the comparison and one limitation.

Work for 5 minutes. Be ready to explain your denominators.

Group Exercise Data

Student	Sleep hours	High stress?	Student	Sleep hours	High stress?
S01	5.0	Yes	S07	7.0	Yes
S02	5.5	Yes	S08	7.0	No
S03	6.0	Yes	S09	7.5	No
S04	6.0	No	S10	8.0	No
S05	6.5	Yes	S11	8.0	No
S06	6.5	No	S12	9.0	No

Report: overall stress counts and percentages; shorter-sleep percentages within each stress group; one comparison and one limitation.

Group Exercise Debrief

Overall: **5 of 12 (41.7%)** report high stress.

High stress reported	Less than 7 hours within the group
Yes	4/5 = **80.0%**
No	2/7 = **28.6%**

In this fictional sample, shorter sleep is more common among students reporting high stress.

How Would We Summarize Sleep Hours?

We have described categories using **counts and percentages**.

What should we ask about the continuous sleep measurements to summarize them?

- What is a typical value? **Center**
- How different are the values? **Spread**
- What does the distribution look like? **Shape**

Review: Mean, Median, and Mode

Sorted sleep hours:

5, 5.5, 6, 6, 6.5, 6.5, 7, 7, 7.5, 8, 8, 9

Calculate the mean, median, and mode.

Mean: total hours ÷ number of students = **6.83 hours**

Median: average of the two middle values = **6.75 hours**

Modes: 6, 6.5, 7, and 8 hours — each appears twice.

The Same Mean Can Hide Different Spread

Fictional group	Sleep hours	Mean	Range
A	6, 7, 8	7	
B	4, 7, 10	7	

Calculate the range for each group. Which group has more spread?

The Same Mean Can Hide Different Spread

Fictional group	Sleep hours	Mean	Range
A	6, 7, 8	7	2
B	4, 7, 10	7	6

Would “both groups average 7 hours” tell the whole story?

Connection to Prep 3

For your assigned dataset, you will:

- Identify your variables and their types.
- Summarize your continuous outcome with **n, mean, SD, minimum, and maximum**.
- Describe a histogram and, after the correlation lesson, a scatterplot.

Next time: continuous summaries, graphs, and Excel practice.

Exit Check

1. One difference between a **sample statistic** and a **population parameter**.
2. A suitable summary for a **yes/no variable**.
3. What you must identify before calculating a **percentage**.

Sources

- Yan Zhang. *Statistics Overview and Descriptive Statistics*, APHS 20203, 50-slide PowerPoint supplied with the course materials. Selected concepts and activities adapted for this lesson.
- *Friis' Epidemiology 101*, third edition, Chapter 2: "Epidemiology and Data Presentation."
- Course materials: approved October 20, 2026 outline and *Group Presentation – Prep 3*.

All student examples in this lesson are **fictional teaching examples**.